



Original Article

A pilot, parallel-group, blinded, placebo-controlled, randomised, pragmatic clinical trial investigating the Effect of temazepam on objective and subjective measures of sleep in critically ill patients (the DREAM trial)

Laurie Showler, MBChB ^{a, b, *}, Yasmine Ali Abdelhamid, MBBS, PhD ^{a, b, **},
Melissa J. Ankravs, BPharm, MClinPharm ^{a, b, c}, Jeremy Goldin, MBBS, MM ^d,
Mark P. Plummer, MBBS, PhD ^{e, f}, Brianna Tascone, BBiomed (Hons) ^b, Kathleen Byrne,
RN, MNSc ^b, Andrew Perkins, BSc, RPSGT ^d, Kirk Kee, MBBS, PhD ^{d, g}, Cara Moore, MBBS ^b,
Barry Johnston, MB, BCh, BAO, MBioethics ^b, Jeffrey Presneill, MBBS, MBiostat, PhD ^{a, b},
Adam M. Deane, MBBS, PhD ^{a, b}

^a Department of Critical Care, Melbourne Medical School, University of Melbourne, Melbourne, Cnr Grattan Street & Royal Parade, Parkville, Melbourne, Victoria, Australia; ^b Department of Intensive Care, The Royal Melbourne Hospital, 300 Grattan Street, Parkville, Melbourne, Victoria, Australia;

^c Department of Pharmacy, The Royal Melbourne Hospital, 300 Grattan Street, Parkville, Melbourne, Victoria, Australia; ^d Department of Respiratory and Sleep Disorders Medicine, The Royal Melbourne Hospital, 300 Grattan Street, Parkville, Melbourne, Victoria, Australia; ^e Department of Intensive Care, Royal Adelaide Hospital, Port Road, Adelaide, South Australia, Australia; ^f Adelaide Medical School, The University of Adelaide, 4 North Terrace, Adelaide, South Australia, Australia; ^g The School of Translational Medicine, Monash University, 99 Commercial Road, Melbourne, Victoria, Australia

ARTICLE INFORMATION

Article history:

Received 10 December 2025

Received in revised form

20 January 2026

Accepted 21 January 2026

Keywords:

Benzodiazepines

Critical illness

Intensive care unit

Sleep

Sleep-wake disorders

ABSTRACT

Objective: Patients in the intensive care unit (ICU) suffer from disturbed sleep and pharmacological sleep aids are frequently prescribed despite limited data on their efficacy. The objective of this study was to assess the effect of a single nocturnal dose of the benzodiazepine temazepam on sleep duration and quality in ICU patients.

Design: Prospective, single-centre, blinded, placebo-controlled, parallel-group, randomised clinical trial.

Setting: A tertiary ICU in Australia.

Participants: Adult ICU patients whose treating clinician considered that a pharmacological sleep aid was indicated.

Interventions: A single weight- and age-adjusted dose of temazepam (10–30 mg) or a matching placebo was administered enterally at 21:00 h.

Main outcome measures: The primary outcome was total sleep time between 21:00 and 07:00 h by hourly structured nurse assessment. Secondary outcomes included the evaluation of sleep quality, independently determined by the bedside nurse and patient using the Richards-Campbell Sleep Questionnaire.

Results: Between October 2020 and May 2024, 56 patients received temazepam ($n = 28$) or placebo ($n = 28$). The mean (standard deviation) total sleep time with temazepam was 349 (120) vs. placebo 291 (124) minutes; difference = 57 min (95% confidence intervals: –11 to 130); $p = 0.10$. No differences in total Richards-Campbell Sleep Questionnaire sleep quality were observed when assessed by the nurse (57 (17) vs. 49 (23), $p = 0.15$) or by the patient (50 (28) vs. 51 (23), $p = 0.70$).

Abbreviations: CAM-ICU, Confusion Assessment Method in the Intensive Care Unit; EMG, Electromyogram; Kg, Kilograms; ICU, Intensive care unit; LOS, Length of stay; mcg, Micrograms; mins, Minutes; mg, Milligrams; ml, Millilitres; n, Number; RCSQ, Richards Campbell Sleep Questionnaire; SD, Standard deviation; TST, Total sleep time; 95% CI, 95% confidence intervals.

* Corresponding author at: Intensive Care Unit, The Royal Melbourne Hospital, 300 Grattan Street, Parkville, Victoria 3050, Australia. Tel.: +61 416 958 489; fax: +61 03 9342 8812.

** Corresponding author at: Department of Intensive Care, The Royal Melbourne Hospital, 300 Grattan Street, Parkville, Melbourne, Victoria, Australia.

E-mail addresses: laurie.showler@student.unimelb.edu.au (L. Showler), yasmine.aliabdelhamid@mh.org.au (Y. Ali Abdelhamid).

<https://doi.org/10.1016/j.ccrj.2026.100169>

1441-2772/© 2026 The Authors. Published by Elsevier B.V. on behalf of College of Intensive Care Medicine of Australia and New Zealand. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Conclusion: A single dose of temazepam was not observed to improve the duration or quality of nocturnal sleep for patients in the ICU.

Trial registration: Retrospectively registered with the Australian and New Zealand Clinical Trials Registry on 11th June 2021 (ACTRN 12621000742875).

© 2026 The Authors. Published by Elsevier B.V. on behalf of College of Intensive Care Medicine of Australia and New Zealand. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Most patients admitted to the intensive care unit (ICU) suffer from disordered sleep. When using polysomnography, disordered sleep is characterised by frequent awakenings, a predominance of light sleep, increased daytime sleep and evidence of atypical sleep architecture.^{1,2} The causes for these sleep disturbances are multifactorial and appear to be part of a broader circadian misalignment.^{1–3}

Most critically ill patients rate their sleep in the ICU as poor and report significant distress related to this perturbation.^{4,5} Poor sleep may contribute to an increased risk of delirium, delay liberation from mechanical ventilation, and worsen clinical outcomes.^{5–9} Disturbed sleep in the ICU may also have longer-term ramifications, with increased anxiety, lower quality of life, and persistent circadian dysfunction in ICU survivors 12 months after discharge.^{10–12}

Recent observational data indicate that a pharmacological sleep aid—a medication prescribed with the intent to improve sleep—is administered to between 9% and 17% of patients in Australasian ICUs, and temazepam is amongst the most frequently prescribed drugs for this purpose.^{13,14} Temazepam is an intermediate-acting, enterally administered benzodiazepine that is primarily indicated for the short-term treatment of insomnia and has been demonstrated to improve total sleep time (TST), sleep latency, and subjective sleep quality in the outpatient setting.^{15–17} However, the efficacy of temazepam in the ICU population remains uncertain.¹

The objective of this trial was to assess the effect of a single dose of temazepam on the duration and quality of nocturnal sleep in critically ill patients in the ICU.

2. Methods

This was a prospective, single-centre, blinded, placebo-controlled, parallel-group, randomised clinical trial. The protocol was approved by the Royal Melbourne Hospital's Human Research Ethics Committee (HREC/59713/MH-2019). Registration of the trial with the Australian New Zealand Clinical Trials Registry (ACTRN 12621000742875) was not completed and approved until June 2021, at which point eight patients had been enrolled. No changes to the study protocol occurred during the study period. There was no external funding.

2.1. Setting

This trial was conducted in a mixed medical, surgical, and trauma ICU.¹⁸ All patients were cared for in single rooms separated by electrochromic glass that provides partial isolation from external noise and light. Twenty-two rooms have direct exposure to daylight, and 16 rooms have indirect or artificial illumination. Overhead lights in the unit were dimmed between 21:00 and 07:00 h. The lighting in the patient's immediate bed area was determined by the bedside care team. The use of earplugs and eye masks was not protocolised and did not occur routinely.

2.2. Patients

Screening was performed on weekdays when all required staff (study intensivist, pharmacist, and researcher) were available. On these days, all patients admitted to the ICU were screened for eligibility. Critically ill patients aged 18 years or older whose treating clinician considered were likely to remain in ICU that evening and that a pharmacological sleep aid was indicated were assessed for inclusion. Patients who were pregnant, those with known hypersensitivity to temazepam, or those in whom the treating clinician believed that an alternative sedating drug was in the best interest of the patient were excluded. Once eligibility was determined, written informed consent was obtained from either the patient or their medical treatment decision maker.

2.3. Randomisation and blinding

The randomisation schedule was a computer-generated 1:1 schedule using the Stata routine *ralloc* function (Stata version 15.1, College Station, Texas), implemented using the secure, web-based software platform, REDCap.^{19,20} The randomisation schedule was securely stored separate from the investigators. The patient, treating healthcare workers, researchers, and statisticians were unaware of treatment allocation until after the statistical analysis was complete.

2.4. Study drug

The dose of temazepam was approximately 0.3 mg/kg, with a reduction in older patients.^{21,22} Accordingly, patients were allocated to one of three doses according to age and body weight: 10 mg (actual body weight <50 kg and/or age >80 years), 20 mg (actual body weight ≤100 kg and age <80 years), or 30 mg (actual body weight >100 kg and age ≤80 years). The dose could be reduced at the treating clinician's discretion. An independent pharmacist or research nurse was responsible for preparing the temazepam in a syringe with 10 ml of water or a matching placebo. This pharmacist or nurse was not involved in ongoing patient care or outcome assessment. Syringes were covered in opaque foil and stored in a locked drug cupboard prior to administration.

The intervention was administered by the patient's nurse at 21:00 h on the study day via an enteral feeding tube or by mouth as appropriate. The syringe was retained overnight for study drug reconciliation the following morning. The intervention was administered for a single night only. The prescription of any new potentially sedative medications was not permitted on the study evening.

2.5. Study outcomes

The primary outcome was TST between 21:00 and 07:00 h, determined by structured nurse assessment as used in a previous drug trial evaluating sleep in the ICU.²³ Patients were evaluated

hourly to determine sleep status over the preceding 60 min and summed for the 10-h period.²³

Secondary outcomes included measurement of the patient's sleep quality on the morning following the intervention using the Richards-Campbell Sleep Questionnaire (RCSQ), which was completed by the bedside nurse and independently by the patient, when appropriate.^{24–26} The RCSQ has been validated against polysomnography and utilises visual analogue scales to measure five domains of sleep quality.^{27,28} The total RCSQ score, calculated as the mean of the subscale scores, summarises sleep quality, where higher scores indicate better sleep quality.²⁹ The prevalence of delirium at 20:00 h on the evening before and 08:00 h on the morning after study drug administration was measured by the bedside nurse using the Confusion Assessment Method in the ICU tool (CAM-ICU).³⁰ Hospital mortality and length of stay data were collected. The use of all medications, including those with potential sedative-hypnotic effects, during the study period and preceding 12 h was recorded. Polysomnography was conducted in a small subgroup.³¹

2.6. Adverse events

Adverse events were defined and reported as detailed in the supplemental material.³²

3. Statistical analysis

Statistical analysis was conducted using Stata (version 19, College Station, Texas), unless otherwise reported. Numeric data were summarised as mean (standard deviation [SD]) or count with an accompanying proportion of a relevant total. Continuous variables were initially compared using two-sided t-tests, while dichotomous variables were compared using Chi-squared tests. The prespecified primary intention-to-treat outcome was nurse-observed TST. This was analysed using a simple univariable linear model containing only a term for randomised treatment group. Two sensitivity analyses of the primary outcome were conducted by applying a multivariable linear model, firstly to the observed data and secondly to a full dataset following multiple imputation of missing TST observations. Both exploratory multivariable linear models included the following terms: sex, patient age, admission category, severity of illness, and delirium status prior to receiving the study drug. The multiple imputation process used truncated regression in 20 sets of chained equations.

The Bayesian Regression Models using Stan (BRMS) package in the statistical software R (version 4.5.0) was used to perform a post-hoc, exploratory Bayesian analysis using 800,000 simulations (400 chains of 2000) with uninformative priors on a multiply imputed dataset.^{33,34} Missing values for sleep duration were generated with 100 runs to return a single set of pooled estimates.³⁵

3.1. Sample size

Based on a mean (SD) TST of 180 (80) minutes in patients receiving usual care, 56 patients were predicted to provide 80% power to detect a difference of 60 min at a two-sided alpha of 0.05.^{23,36}

For logistical reasons, the study drug was dispensed during business hours. Because patients could be discharged after hours (i.e. after study drug was dispensed but before it was administered), patients who were discharged or discovered to meet an exclusion criterion prior to study drug administration were withdrawn without knowledge of their treatment allocation and were not analysed. If a participant remained in ICU but refused to take the study drug once dispensed, they were included in the

intention-to-treat analysis. Randomisation was allowed to continue until an accumulated total of 56 evaluable patients was achieved.

4. Results

4.1. Participants

Between October 2020 and May 2024, a total of 60 patients were enrolled and randomised. Four patients were withdrawn before the study drug was administered, leaving the target of 56 evaluable patients (Fig. 1).

4.2. Patient characteristics

Baseline demographics of participants in the temazepam and placebo groups were comparable except for admission category and admission reason (Table 1). The use of potentially sedating medications in the 12 h preceding and 12 h after study drug administration was similar (Supplemental Table 1).

4.3. Compliance with study drug

The assigned intervention was administered to 54 of 56 patients who were dispensed the study drug (28 (100%) temazepam, 26 (93%) placebo). Reasons for noncompliance were patient refusal and delirium. The overall mean (SD) dose of temazepam administered was 21 (5.4) mg (Supplemental Fig. 1), administered by enteral feeding tube for 13 (46%) patients and 12 (42%) patients in the temazepam and placebo groups, respectively. The study drug was administered within an hour of its intended administration time in 23 (82%) patients for both groups (Supplemental Fig. 2).

4.4. Primary outcome

Data on nurse-observed TST were available for 53 patients (26 (95%) temazepam, 27 (96%) placebo). The mean (SD) nurse-observed TST was 349 (120) minutes for patients receiving temazepam and 291 (124) minutes for those receiving placebo; difference = 57 min, (95% confidence intervals [CI]: –11 to 130); $p = 0.10$ (Fig. 2A).

The primary outcome was not available for three patients. Two patients refused study drug (1 (3.6%) temazepam, 1 (3.6%) placebo), and one patient's sleep duration was not recorded (1 (3.6%) temazepam).

Sensitivity analyses derived from multivariable linear models applied to the observed and a multiply imputed dataset are summarised in Table 2 and Supplemental Tables 2, 3 and 4.

The exploratory Bayesian analysis using imputed data found a 97% posterior probability that temazepam was associated with some increase in sleep duration, with a 60% posterior probability that there was at least 60 min of additional sleep and an 8% posterior probability that it was greater than 120 min (Supplemental Fig. 3).

4.5. Secondary outcomes

Data for nurse-reported RCSQ were available for 51 patients (25 (89%) temazepam, 26 (95%) placebo). The mean (SD) nurse-reported total RCSQ was 57 (16) for patients receiving temazepam and 49 (22) for those receiving placebo; difference = 8.1 (95% CI: –3.0 to 19); $p = 0.15$ (Fig. 2B). One nurse-reported RCSQ was missing due to patient refusal of study drug (1 (3.6%) placebo), and the RCSQ was not recorded in the remaining four patients (3 (8.3%) temazepam, 1 (3.6%) placebo).

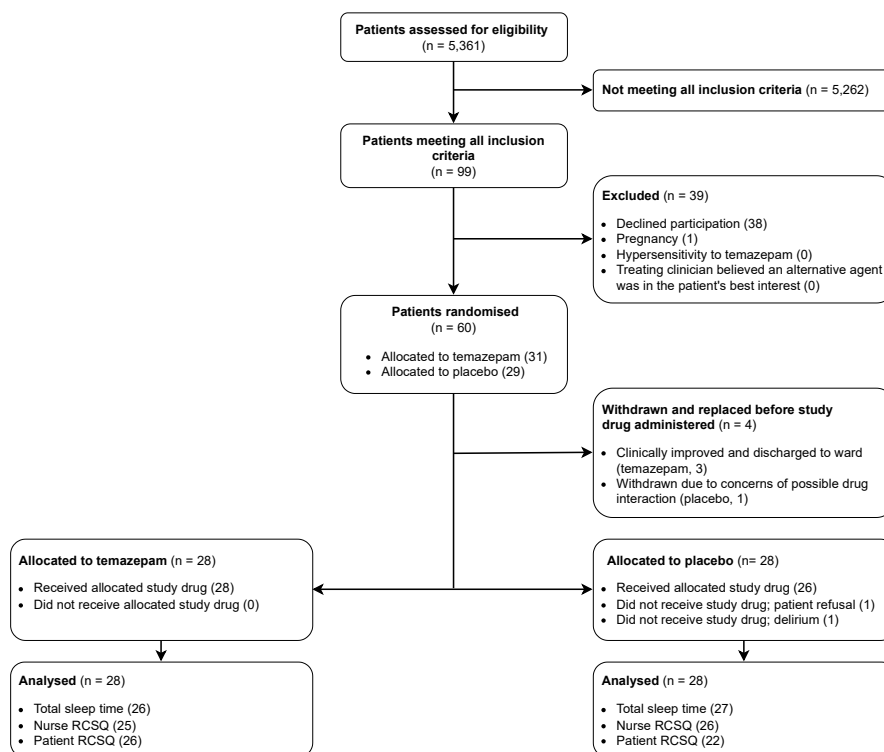


Fig. 1. CONSORT diagram.

RCSQ, Richards-Campbell sleep questionnaire; n, number.

Data for patient-reported RCSQ were available for 48 patients (26 (95%) temazepam, 22 (84%) placebo). The mean (SD) patient-reported total RCSQ was 54 (24) for patients receiving temazepam and 51 (23) for those receiving placebo; difference = 2.7 (95% CI: -11 to 17); $p = 0.69$ (Fig. 2C). One patient-reported RCSQ was missing due to refusal of study drug (1 (3.6%) placebo), one patient declined to provide an answer (1 (3.6%) placebo), and six were unable to complete the assessment (2 (7.1%) temazepam, 4 (14.2%) placebo).

Subcategory scores for nurse-completed RCSQ showed no significant differences between groups, except for mean (SD) sleep latency with temazepam 68 (19) compared to placebo 49 (25); difference = 19 (95% CI: 6.6 to 32), $p = 0.004$. Subcategory scores for patient-completed RCSQ showed no significant differences between groups (Supplemental Tables 5 and 6).

Linear regression of patient-reported total RCSQ and nurse-reported total RCSQ identified no strong explanatory variables beyond treatment allocation. The correlation of patient total RCSQ and nurse total RCSQ was 0.53 (95% CI: 0.21 to 0.85) (Fig. 3).

4.6. Polysomnography

Polysomnography was completed in 6 patients (4 (7.0%) temazepam, 2 (3.5%) placebo). Baseline characteristics for this subgroup and outcomes are provided in Supplemental Tables 7 and 8

4.7. Delirium

Confusion Assessment Method-ICU assessments were completed before and after study drug administration for all patients. There was no statistically significant between-group difference for the incidence of delirium (Table 3).

4.8. Mortality, length of stay, and adverse events

Mortality and length of stay data are summarised in Table 3. One adverse event was recorded, involving hypotension requiring vasopressor support that occurred 30 min after administration of temazepam.

5. Discussion

5.1. Summary of results

The main finding is that a single dose of enterally administered temazepam probably increased the duration of patients' sleep. While the observed difference did not meet the threshold for statistical significance using a frequentist approach, Bayesian analyses provided strong support for longer sleep duration following temazepam. Regardless of treatment allocation, patients spent a large proportion of the available nocturnal sleep period awake. No between-group differences in sleep quality were identified when assessed by the nurse or the patient. The incidence of delirium on the evening prior to and the morning after study drug administration was low and not statistically different between groups. Adverse events were infrequent.

5.2. Comparison to other trial data using pharmacological sleep aids in ICU

This is the first published study of the effect of an enterally administered benzodiazepine on sleep duration and quality in patients admitted to an ICU. However, several other enterally administered sedative-hypnotics have been evaluated.³⁷ Melatonin remains the most extensively studied enteral medication for

Table 1
Comparison of baseline characteristics.

	Temazepam (n = 28)	Placebo (n = 28)	Both (n = 56)
Age, years	55.7 (16.7)	62.1 (14.6)	58.9 (15.8)
Female sex, n (%)	8 (29)	5 (18)	13 (23)
Weight, kilograms	87.5 (30.1)	102.3 (37.2)	94.9 (34.6)
Day of ICU admission	10.1 (11.1)	6.6 (6.9)	8.3 (9.4)
Admission category, n (%)			
- Operative (elective)	2 (7.1)	3 (10.7)	5 (8.9)
- Operative (emergency)	4 (14.3)	11 (39.3)	15 (26.8)
- Nonoperative	22 (78.6)	14 (50.0)	36 (64.3)
Admission source, n (%)			
- Emergency	10 (35.7)	9 (32.1)	19 (33.9)
- Ward	9 (32.1)	6 (21.4)	15 (26.8)
- Other hospital	4 (14.3)	1 (3.6)	5 (8.9)
- Other ICU	0 (0.0)	1 (3.6)	1 (1.8)
- Theatre (elective)	2 (7.1)	3 (10.7)	5 (8.9)
- Theatre (emergency)	3 (10.7)	8 (28.6)	11 (19.6)
Admission reason, n (%)			
- Trauma	5 (17.9)	3 (10.7)	8 (14.3)
- Pneumonia/ARDS	5 (17.9)	4 (14.3)	9 (16.1)
- Sepsis	6 (21.4)	1 (3.6)	7 (12.5)
- Cardiogenic shock	1 (3.6)	0 (0.0)	1 (1.8)
- Neurological	2 (7.1)	0 (0.0)	2 (3.6)
- Other	9 (32.1)	20 (71.4)	29 (51.8)
APACHE III score	64.0 (18.7)	60.9 (21.3)	62.4 (20.1)
RASS score	0 (0.4)	0 (0.7)	0 (0.5)
ICU interventions, n (%)			
- Tracheostomy	8 (28.6)	4 (14.3)	12 (21.4)
- Invasive ventilation	5 (19.2)	2 (7.1)	7 (12.5)
- Renal replacement	2 (7.1)	1 (3.6)	3 (5.4)
- Vasopressors	7 (25.0)	6 (21.4)	13 (23.2)
- Enteral nutrition	12 (42.9)	8 (28.6)	20 (35.7)
Medical history			
- Obstructive sleep apnoea	1 (3.6)	4 (14.3)	5 (8.9)
- Current tobacco use	4 (14.3)	3 (10.7)	7 (12.5)
- Alcohol misuse	3 (10.7)	3 (10.7)	6 (10.7)

Data presented as mean (standard deviation) or number (percentage).

APACHE, acute physiology and chronic health evaluation; ARDS, acute respiratory distress syndrome; ICU, intensive care unit; RASS, Richmond Agitation and Sedation Scale.

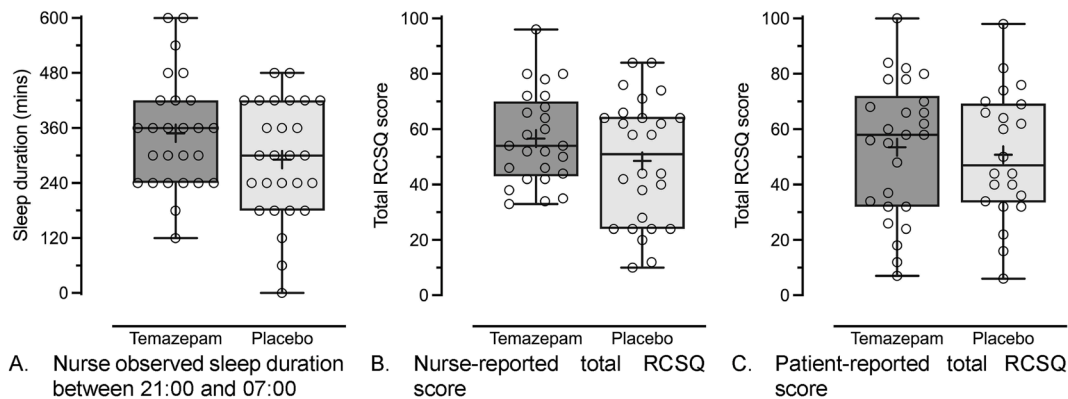


Fig. 2. Box-and-whisker plot demonstrating the differences in nurse-observed sleep duration, nurse-reported RCSQ and patient-reported RCSQ with temazepam and placebo. RCSQ, Richards-Campbell Sleep Questionnaire; mins, minutes.

Whiskers: minimum and maximum values, horizontal bar: median, cross: mean, upper limit of box: 75th centile, lower limit of box: 25th centile, dots: individual data point.

promoting sleep in the ICU and is the only pharmacological sleep aid recommended by professional society guidelines in this setting.³⁸ Of the six randomised clinical trials comparing melatonin to placebo (n = 1180 patients), none have demonstrated a significant improvement in sleep duration, and only one reported a subjective improvement in sleep quality.^{23,39–43} There has been interest in newer medications including the melatonin receptor antagonist ramelteon and the orexin receptor agonist suvorexant, but no difference in sleep quantity or quality has been reported in the available trial data.^{44–47} A randomised, open-label study of 100 mg gabapentin compared with standard management in 60

ICU patients reported statistically significant improvements in both bispectral index-derived TST and quality (total RCSQ) after 5 days of treatment.⁴⁸ However, the open-label design and differences in baseline TST raise uncertainty regarding the true magnitude of the effect size.

5.3. Comparison to data using temazepam outside of the ICU

Data evaluating the effect of temazepam in patients on non-ICU hospital wards are sparse.^{49,50} A blinded randomised clinical trial comparing temazepam, melatonin, and placebo in 20 patients

Table 2
Effect estimates of temazepam on total sleep time.

Total sleep time, minutes	Mean difference with temazepam	95% confidence intervals	p value
Unadjusted (primary outcome)	57	–11 to 130	0.10
Adjusted multivariable linear model			
- Observed data ^a	65	–11 to 140	0.09
- Multiply imputed data ^b	68	–5.4 to 140	0.07

^a Refer to Supplemental Table 2.
^b Refer to Supplemental Table 3.

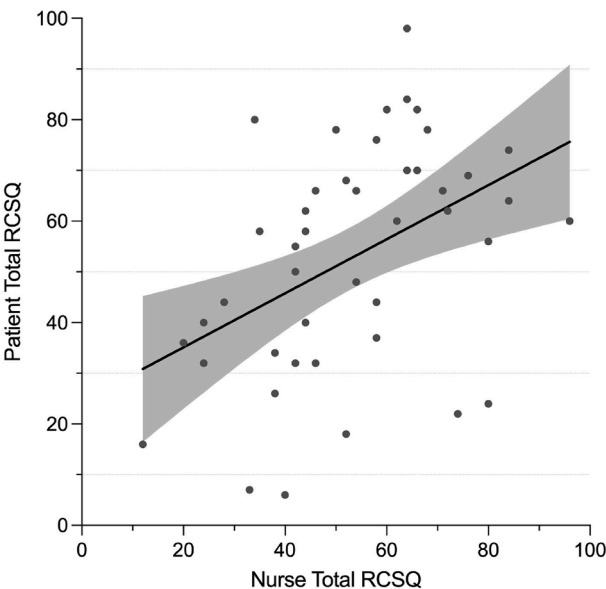


Fig. 3. Scatter plot comparing patient- and nurse-reported total Richards-Campbell Sleep Scores.
RCSQ, Richards-Campbell Sleep Questionnaire.
n = 45, dots: individual RCSQ scores, solid line: superimposed simple linear plot with a coefficient of 0.53 (95% CI: 0.21 to 0.85), shaded area: 95% confidence intervals of regression line.

with advanced cancer included only four in-patients who received temazepam. While temazepam improved subjective sleep quality compared with placebo, inferences for patients in the ICU are limited.

The evidence supporting the use of temazepam for the treatment of chronic insomnia in the community was recently summarised.¹⁵ Meta-analysis of three randomised trials (n = 73 patients) observed that mean sleep latency was reduced by 9 min (95% CI: 4–22 min) and duration of sleep (mean TST) was increased by 99 min (95% CI: 63–135 min) with 15 mg temazepam compared with placebo. Increased doses of temazepam were studied, specifically 20 mg and 30 mg, but a consistent relationship between dose and TST was not identified.^{36,51–55}

5.4. Strengths and limitations

This trial has several strengths. While frequently prescribed in Australasia, this is the first randomised controlled trial evaluating the impact of temazepam on sleep in the ICU. As there is no single, universal metric to evaluate sleep, the use of objective and subjective measurement tools in this study allowed a broader understanding of pharmacodynamics. There was high compliance with subjective assessment of sleep, which contrasts with studies reporting that up to 48% of ICU patients are unable to engage in self-assessment.²⁵

However, the trial has several important limitations. This was a relatively small sample and may be underpowered to detect significant changes in secondary outcomes, especially sleep quality and delirium. As such, these outcomes should be interpreted cautiously. As a single-centre study conducted in an Australian ICU, differences in staffing ratios, ICU design, sedation practices, and ICU protocols may limit external validity.

Temazepam appears to have a favourable side effect profile, but concerns relating to the adverse effects of benzodiazepine exposure, including an increased incidence of delirium in ICU patients, an observed increase in mortality in outpatient settings, and the risk of dependence, were not able to be adequately assessed in this single-dose, single-centre trial.^{15,56,57} While continuous infusions of benzodiazepines have been linked to an increased risk of delirium, the risk related to intermittent dosing remains less clear.^{58,59} It is possible that a single dose of temazepam could cause delirium many hours or days after administration, and the final delirium assessment, which occurred 11 h poststudy drug administration, would not have captured this. ICU delirium is associated with increased short-term mortality, and exposure to any agent that has the potential to promote delirium may be unfavourable.⁶⁰ As such, potential adverse effects should be weighed against any possible improvements associated with an increase in sleep time.

The optimal dosing regimen and treatment duration for temazepam are not well defined. A weight- and age-adjusted dose was used in this trial, but a clear dose–response relationship between temazepam dose and TST is not established.¹⁵ A single-night intervention was chosen for pragmatic reasons, but repeated dosing over successive nights could have potentially consolidated the effect of temazepam on sleep duration and circadian alignment or demonstrated evidence of tolerance and tachyphylaxis with increased adverse events.^{61,62}

The structured nurse assessment for TST used in this study has been used previously and was selected because of the pragmatic nature of the trial. However, it has not been validated against polysomnography.²³ An alternative method, the Edwards-Schuring Sleep Observation Tool, has undergone such validation but requires patient assessments every 15 min.⁶³ In the study ICU, it was impractical to add this frequency of recording to existing busy nursing workloads. Structured nurse assessment may be prone to bias and be insufficiently sensitive or specific to accurately determine a patient's sleep state in the presence of concurrent sedation or an altered conscious state from other causes, resulting in an overestimation of TST.⁶⁴ The poor agreement between TST and subjective sleep quality in this study may be attributable to misclassification of sedation as sleep. However, this discordant relationship has previously been described and postulated to be due to atypical sleep, persisting sleep deprivation, or that these two measures address different aspects of sleep.^{64–66} While concurrent use of polysomnography was planned and would have permitted validation of the nurse assessment, polysomnography recording for all participants was limited by logistical issues and patient preference.

Table 3

Table comparing length of stay, hospital mortality, adverse events, and delirium for patients receiving temazepam and placebo.

	Temazepam (n = 28)	Placebo (n = 28)	p-value
Length of stay poststudy drug administration			
- ICU LOS, <i>day</i>	6.5 (9.1)	3.7 (4.1)	0.16
- Hospital LOS, <i>days</i>	17.4 (10.5)	21.4 (25.3)	0.45
Hospital mortality, <i>n (%)</i>	5 (17.9)	2 (7.1)	0.23
Adverse events, <i>n (%)</i>	1 (3.6)	0 (0.0)	0.31
Delirium, <i>n (%)</i>			
- CAM ICU positive prestudy	3 (10.7)	5 (17.9)	0.45 ^a
- CAM ICU positive poststudy	1 (3.6)	4 (14.3)	0.16 ^a

Data presented as mean (standard deviation) or number (percentage).

ICU, intensive care unit; n, number; CAM-ICU, Confusion Assessment Method for ICU; LOS, length of stay.

^a Chi-square.

Patients in the ICU frequently do not have a single consolidated nocturnal rest period, and instead, their sleep is spread over 24 h.⁶⁷ Consequently, by limiting our assessment of sleep to just the night-time period, the full effect of the intervention on patients' sleep over the entire 24-h period may not have been fully appreciated.

This study enrolled 60 patients over 44 months, including a pause in enrolment for four months during 2021 due to logistical limitations imposed by a surge in COVID-19 patients in Melbourne. Nonetheless, recruitment was slower than expected, with point-prevalence data indicating that a pharmacological sleep aid is prescribed to around 10% of patients in Australasian ICUs.¹⁴ The cause for this slow rate of recruitment is unknown but likely multifactorial. Screening was restricted to weekdays when all staff (study intensivist, pharmacist, and researcher) were available, and due to competing demands and leave, screening did not occur every day. Potential patients required that a pharmacological sleep aid be indicated, the treating clinician had no preference for a particular drug, and the patient was competent to provide written informed consent, or this could be obtained from their medical treatment decision maker. We did not obtain consent in 38 of the 99 patients approached, and the refusal rate was greater than we anticipated.

5.5. Future directions and clinical implications

As a preliminary, single-centre trial, with some missing data, the observed trend towards longer nurse-observed TST, improved nurse-reported sleep quality, and low rate of adverse events seen in those receiving temazepam raises the prospect that a larger, multicentre study may be justified to fully define the effects of temazepam or other similar benzodiazepine receptor agonists. Reproducing this study with a placebo-control may not be possible in some jurisdictions, and the use of an appropriate active control may be preferable. However, the absence of any patient-reported improvements in sleep quality and broader concerns relating to the potential adverse effects of benzodiazepine use potentially suggests that future research on alternative agents might yield more positive results. Such agents include clonidine, gabapentin or other novel therapies.^{48,68–70}

Given the multifactorial nature of sleep disruption in the ICU population, in addition to the use of sleep-promoting medications, a more holistic approach to optimising sleep and circadian rhythms may be necessary.^{30,69–71} Future studies should include both objective and subjective measures of sleep quantity and quality to facilitate a full understanding of any intervention.

The structured nurse assessment used to determine TST provided a pragmatic and simple means to measure sleep, but the validation of this against polysomnography is a priority for future research into sleep in the critically ill. The low number of polysomnography performed during this study highlights the challenges of using this technique in the ICU setting.⁷²

6. Conclusion

The administration of temazepam in ICU patients did not increase nurse-observed total nocturnal sleep duration or subjective sleep quality. While observed changes in sleep duration did not reach statistical significance, sensitivity analyses provide support for a larger study. Nonetheless, the absence of improvement in any patient-oriented outcome suggests that clinicians should be circumspect in prescribing temazepam as a pharmacological sleep aid outside of a clinical trial.

CRediT authorship contribution statement

LS Formal analysis, Methodology, Project administration, Supervision, Visualisation, Writing – Original Draft Preparation.

YA Conceptualisation, Methodology, Project Administration, Supervision, Writing – Review & Editing.

MA Conceptualisation, Methodology, Project Administration, Writing – Review & Editing.

JG Writing – Review & Editing.

MP Conceptualisation, Methodology, Project Administration, Writing – Review & Editing.

BT Conceptualisation, Data Curation, Investigation, Methodology, Project Administration.

KB Conceptualisation, Investigation, Methodology, Project Administration, Writing – Review & Editing.

AP Investigation, Methodology, Project Administration, Writing – Review & Editing.

KK Conceptualisation, Methodology, Writing – Review & Editing.

CM Conceptualisation, Methodology, Writing – Review & Editing.

BJ Conceptualisation, Methodology, Writing – Review & Editing.

JP Conceptualisation, Formal analysis, Methodology, Visualisation, Writing – Review & Editing.

AD Conceptualisation, Funding Acquisition, Methodology, Project Administration, Supervision, Writing – Review & Editing.

Consent for publication

Not applicable.

Ethics approval and consent to participate

The protocol was approved by the Melbourne Health Research Ethics Committee (HREC/59713/MH-2019). Prior to enrolment into the study, informed consent was obtained from the patient or their legally delegated Medical Treatment Decision Maker.

Funding

This study was supported by departmental funding. No external funding was sought.

Conflict of interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Given her role as Associate Editor for Critical Care and Resuscitation, Yasmine Ali Abdelhamid, had no role in the peer review of this article and had no access to information regarding its peer review, full responsibility for the editorial process for this article was delegated to another journal editor. Given his role as Associate Editor for Critical Care and Resuscitation, Mark Plummer, had no role in the peer review of this article and had no access to information regarding its peer review, full responsibility for the editorial process for this article was delegated to another journal editor. Given his role in the Editorial Board for Critical Care and Resuscitation, Adam Deane, had no role in the peer review of this article and had no access to information regarding its peer review, full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank Lewis Hackenberger (Royal Melbourne Hospital, Intensive Care Unit, Data Report Lead) and the Royal Melbourne Hospital's Intensive Care Data team for their assistance with data collection, and Lucy Sharrock and Megan Richardson (Royal Melbourne Hospital, Pharmacy Department) for their assistance with study drug preparation. We also thank Briannah Miles (Royal Melbourne Hospital, Intensive Care Unit, Research Scientist) for project administration and investigation.

Laurie Showler would like to acknowledge the support provided by the Australian Commonwealth Government through an Australian Government Research Training Program Scholarship.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ccrj.2026.100169>.

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

References

- [1] Showler L, Ali Abdelhamid Y, Goldin J, Deane AM. Sleep during and following critical illness: a narrative review. *World J Crit Care Med* 2023;12(3):92–115. <https://doi.org/10.5492/wjccm.v12.i3.92> [PMID: 37397589].
- [2] Telias I, Wilcox ME. Sleep and circadian rhythm in critical illness. *Crit Care* 2019;23(1):82. <https://doi.org/10.1186/s13054-019-2366-0> [PMID: 30850003].
- [3] Knauer MP, Ayas NT, Bosma KJ, Drouot X, Heavner MS, Owens RL, et al. Causes, consequences, and treatments of sleep and circadian disruption in the ICU: an official American thoracic society research statement. *Am J Respir Crit Care Med* 2023;207(7):e49–68. <https://doi.org/10.1164/rccm.202301-0184ST> [PMID: 36999950].
- [4] Little A, Ethier C, Ayas N, Thanachayanont T, Jiang D, Mehta S. A patient survey of sleep quality in the intensive care unit. *Minerva Anestesiol* 2012;78(4):406–14 [PMID: 22337154].
- [5] Simini B. Patients' perceptions of intensive care. *Lancet* 1999;354(9178):571–2. [https://doi.org/10.1016/S0140-6736\(99\)02728-2](https://doi.org/10.1016/S0140-6736(99)02728-2) [PMID: 10470711].
- [6] Cabello B, Thille AW, Drouot X, Galia F, Mancebo J, d'Ortho MP, et al. Sleep quality in mechanically ventilated patients: comparison of three ventilatory modes. *Crit Care Med* 2008;36(6):1749–55. <https://doi.org/10.1097/CCM.0b013e3181743f41> [PMID: 18496373].
- [7] Boyko Y, Toft P, Ørding H, Lauridsen JT, Nikolic M, Jennum P. Atypical sleep in critically ill patients on mechanical ventilation is associated with increased mortality. *Sleep Breath* 2019;23(1):379–88. <https://doi.org/10.1007/s11325-018-1718-3> [PMID: 30215172].
- [8] Dres M, Younes M, Rittayamai N, Kendzerska T, Telias I, Grieco DL, et al. Sleep and pathological wakefulness at the time of liberation from mechanical ventilation (SLEEVE). A prospective multicenter physiological study. *Am J Respir Crit Care Med* 2019;199(9):1106–15. <https://doi.org/10.1164/rccm.201811-2119OC> [PMID: 30818966].
- [9] Daou M, Telias I, Younes M, Brochard L, Wilcox ME. Abnormal sleep, circadian rhythm disruption, and delirium in the ICU: are they related? *Front Neurol* 2020;11. <https://doi.org/10.3389/fneur.2020.549908> [PMID: 33071941].
- [10] Henriquez-Beltrán M, Vaca R, Benítez ID, González J, Santistev S, Aguilá M, et al. Sleep and circadian health of critical survivors: a 12-Month Follow-Up study. *Crit Care Med* 2024;52(8):1206–17. <https://doi.org/10.1097/ccm.0000000000006298> [PMID: 38597721].
- [11] Altman MT, Knauer MP, Pisani MA. Sleep disturbance after hospitalization and critical illness: a systematic review. *Ann Am Thorac Soc* 2017;14(9):1457–68. <https://doi.org/10.1513/AnnalsATS.201702-148SR> [PMID: 28644698].
- [12] Orwelius L, Nordlund A, Nordlund P, Edell-Gustafsson U, Sjöberg F. Prevalence of sleep disturbances and long-term reduced health-related quality of life after critical care: a prospective multicenter cohort study. *Crit Care* 2008;12(4):R97. <https://doi.org/10.1186/cc6973> [PMID: 18673569].
- [13] Wong C, Ho J, Ankrav MJ, Sharrock L, Kee K, Goldin J, et al. Administration of pharmacological sleep aids prior to, during and following critical illness. *Intern Med J* 2022;52(11):1962–70. <https://doi.org/10.1111/imj.15492> [PMID: 34392601].
- [14] Showler L, Deane AM, Litton E, Ankrav MJ, Wibrow B, Barge D, et al. A multicentre point prevalence study of nocturnal hours awake and enteral pharmacological sleep aids in patients admitted to Australian and New Zealand intensive care units. *Crit Care Resusc* 2024;26(3):192–7. <https://doi.org/10.1016/j.ccrj.2024.06.009> [PMID:].
- [15] Sateia MJ, Buysse DJ, Krystal AD, Neubauer DN, Heald JL. Clinical practice guideline for the pharmacologic treatment of chronic insomnia in adults: an American academy of sleep medicine clinical practice guideline. *J Clin Sleep Med* 2017;13(2):307–49. <https://doi.org/10.5664/jcsm.6470> [PMID:].
- [16] Chiu H-Y, Lee H-C, Liu J-W, Hua S-J, Chen P-Y, Tsai P-S, et al. Comparative efficacy and safety of hypnotics for insomnia in older adults: a systematic review and network meta-analysis. *Sleep* 2020;44(5). <https://doi.org/10.1093/sleep/zsaa260> [PMID:].
- [17] Heel RC, Brogden RN, Speight TM, Avery GS. Temazepam: a review of its pharmacological properties and therapeutic efficacy as an hypnotic. *Drugs* 1981;21(5):321–40. <https://doi.org/10.2165/00003495-198121050-00001> [PMID: 6112127].
- [18] Northcott MH, Johnston G, Presneill JJ, Fazio TN, Adamson N, Ankrav MJ, et al. Aggression, violence and threatening behaviour during critical illness. *Crit Care Resusc* 2023;25(2):65–70. <https://doi.org/10.1016/j.ccrj.2023.05.002> [PMID: 37876598].
- [19] Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap)—A metadata-driven methodology and workflow process for providing translational research informatics support. *J Biomed Inf* 2009;42(2):377–81. <https://doi.org/10.1016/j.jbi.2008.08.010> [PMID:].
- [20] Harris PA, Taylor R, Minor BL, Elliott V, Fernandez M, O'Neal L, et al. The REDCap consortium: building an international community of software platform partners. *J Biomed Inf* 2019;95:103208. <https://doi.org/10.1016/j.jbi.2019.103208> [PMID:].
- [21] *Paediatric handbook*. 8th ed. Wiley-Blackwell; 2009.
- [22] Thomson K, Tey D, Marks M. Appendix 2 - pharmacopoeia. In: *Paediatric Handbook*. 8th ed. Oxford (UK): Wiley-Blackwell; 2009. p. 596.
- [23] Ibrahim MG, Bellomo R, Hart GK, Norman TR, Goldsmith D, Bates S, et al. A double-blind placebo-controlled randomised pilot study of nocturnal melatonin in tracheostomised patients. *Crit Care Resusc* 2006;8(3):187–91 [PMID: 16930101].
- [24] Richards KC, O'Sullivan PS, Phillips RL. Measurement of sleep in critically ill patients. *J Nurs Meas* 2000;8(2):131–44 [PMID: 11227580].
- [25] Frisk U, Nordström G. Patients' sleep in an intensive care unit—patients' and nurses' perception. *Intensive Crit Care Nurs* 2003;19(6):342–9. [https://doi.org/10.1016/s0964-3397\(03\)00076-4](https://doi.org/10.1016/s0964-3397(03)00076-4) [PMID: 14637294].
- [26] Kamdar BB, Shah PA, King LM, Kho ME, Zhou X, Colantuoni E, et al. Patient-nurse interrater reliability and agreement of the richards-campbell sleep questionnaire. *Am J Crit Care* 2012;21(4):261–9. <https://doi.org/10.4037/ajcc2012111> [PMID: 22751369].
- [27] Kakar E, Priester M, Wessels P, Slooter AJC, Louter M, van der Jagt M. Sleep assessment in critically ill adults: a systematic review and meta-analysis. *J Crit Care* 2022;71:154102. <https://doi.org/10.1016/j.jccr.2022.154102> [PMID: 35849874].
- [28] Richards K. Techniques for measurement of sleep in critical care. *Focus Crit Care* 1987;14(4):34–40 [PMID: 3650169].

- [29] Elliott R, Axelin A, Richards KC, Vahlberg T, Ritmala-Castren M. Sensitivity and specificity of proposed richards-campbell sleep questionnaire cut-off scores for good quality sleep during an ICU stay. *J Clin Nurs* 2022. <https://doi.org/10.1111/jocn.16348> [PMID: 35570380].
- [30] Wilcox ME, Burry L, Englesakis M, Coman B, Dao M, van Haren FM, et al. Intensive care unit interventions to promote sleep and circadian biology in reducing incident delirium: a scoping review. *Thorax* 2024;79(10):988–97. <https://doi.org/10.1136/thorax-2023-220036> [PMID: 38350730].
- [31] Delaney LJ, Litton E, Huang HC, Lopez V, van Haren FM. The accuracy of simple, feasible alternatives to polysomnography for assessing sleep in intensive care: an observational study. *Aust Crit Care* 2023;36(3):361–9. <https://doi.org/10.1016/j.aucc.2022.02.004> [PMID: 35361553].
- [32] Pharmacovigilance responsibilities of medicine sponsors: australian recommendations and requirements version 3.0. In: Australian government DoHaAC, therapeutic goods administration editor; 2023.
- [33] Team RC. A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing; 2021.
- [34] Bürkner P-C. Brms: an R package for bayesian multilevel models using stan. *J Stat Software* 2017;80(1):1–28. <https://doi.org/10.18637/jss.v080.i01> [PMID:].
- [35] Zhou X, Reiter JP. A note on bayesian inference after multiple imputation. *Am Statistician* 2010;64(2):159–63 [PMID].
- [36] Cuanang JR, Limos L. Treatment of insomnia with temazepam: double-blind, placebo-controlled evaluation. *Clin Ther* 1982;4(5):402–12 [PMID: 6121626].
- [37] Singh G, Nguyen C, Kuschner W. Pharmacologic sleep aids in the intensive care unit: a systematic review. *J Intensive Care Med* 2024;8850666241255345. <https://doi.org/10.1177/08850666241255345> [PMID: 38881385].
- [38] Lewis K, Balas MC, Stollings JL, McNett M, Girard TD, Chanques G, et al. A focused update to the clinical practice guidelines for the prevention and management of pain, anxiety, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med* 2025;53(3):e711–27. <https://doi.org/10.1097/ccm.0000000000006574> [PMID: 00003246-202503000-00017].
- [39] Bellapart J, Appadurai V, Lassi-Smith M, Stuart J, Zappala C, Boots R. Effect of exogenous melatonin administration in critically ill patients on delirium and sleep: a randomized controlled trial. *Crit Care Res Pract* 2020;3951828 [PMID: 33029397].
- [40] Gandolfi JV, Di Bernardo APA, Chanes DAV, Martin DF, Joles VB, Amendola CP, et al. The effects of melatonin supplementation on sleep quality and assessment of the serum melatonin in ICU patients: a randomized controlled trial. *Crit Care Med* 2020;48(12):e1286–93. <https://doi.org/10.1097/ccm.0000000000004690> [PMID: 33048904].
- [41] Bourne RS, Mills GH, Minelli C. Melatonin therapy to improve nocturnal sleep in critically ill patients: encouraging results from a small randomised controlled trial. *Crit Care* 2008;12(2):R52. <https://doi.org/10.1186/cc6871> [PMID: 18423009].
- [42] Mistraletti G, Umbrello M, Sabbatini G, Miori S, Taverna M, Cerri B, et al. Melatonin reduces the need for sedation in ICU patients: a randomized controlled trial. *Minerva Anestesiol* 2015;81(12):1298–310 [PMID: 25969139].
- [43] Wibrow B, Martinez FE, Myers E, Chapman A, Litton E, Ho KM, et al. Prophylactic melatonin for delirium in intensive care (Pro-MEDIC): a randomized controlled trial. *Intensive Care Med* 2022;48(4):414–25. <https://doi.org/10.1007/s00134-022-06638-9> [PMID: 35220473].
- [44] Jaiswal SJ, Bagsic SRS, Takata E, Kamdar BB, Ancoli-Israel S, Owens RL. Actigraphy-based sleep and activity measurements in intensive care unit patients randomized to ramelteon or placebo for delirium prevention. *Sci Rep* 2023;13(1):1450. <https://doi.org/10.1038/s41598-023-28095-0> [PMID: 36702822].
- [45] Nishikimi M, Numaguchi A, Takahashi K, Miyagawa Y, Matsui K, Higashi M, et al. Effect of administration of ramelteon, a melatonin receptor agonist, on the duration of stay in the ICU: a single-center randomized placebo-controlled trial. *Crit Care Med* 2018;46(7):1099–105. <https://doi.org/10.1097/ccm.0000000000003132> [PMID: 29595562].
- [46] Hatta K, Kishi Y, Wada K, Takeuchi T, Ito S, Kurata A, et al. Preventive effects of suvorexant on delirium: a randomized placebo-controlled trial. *J Clin Psychiatry* 2017;78(8):e970–9. <https://doi.org/10.4088/JCP.16m11194> [PMID: 28767209].
- [47] Hatta K, Kishi Y, Wada K, Takeuchi T, Odawara T, Usui C, et al. Preventive effects of ramelteon on delirium: a randomized placebo-controlled trial. *JAMA Psychiatry* 2014;71(4):397–403. <https://doi.org/10.1001/jamapsychiatry.2013.3320> [PMID: 24554232].
- [48] Susantitapong K, Dilokpattanamongkol P, Sutherasan Y, Liamsombut S, Suthisang C. Effects of gabapentin on slow-wave sleep period in critically ill adult patients: a randomized controlled trial. *Clin Transl Sci* 2024;17(5):e13815. <https://doi.org/10.1111/cts.13815> [PMID: 38803031].
- [49] White B, Snyder HS, Patel MVB. Evaluation of medications used for hospitalized patients with sleep disturbances: a frequency analysis and literature review. *J Pharm Pract* 2023;36(1):126–38. <https://doi.org/10.1177/08971900211017857> [PMID: 34096384].
- [50] Kanji S, Mera A, Hutton B, Burry L, Rosenberg E, MacDonald E, et al. Pharmacological interventions to improve sleep in hospitalised adults: a systematic review. *BMJ Open* 2016;6(7):e012108. <https://doi.org/10.1136/bmjopen-2016-012108>.
- [51] Tuk B, Oberyé JJ, Pieters MS, Schoemaker RC, Kemp B, van Gerven J, et al. Pharmacodynamics of temazepam in primary insomnia: assessment of the value of quantitative electroencephalography and saccadic eye movements in predicting improvement of sleep. *Clin Pharmacol Ther* 1997;62(4):444–52. [https://doi.org/10.1016/s0009-9236\(97\)90123-5](https://doi.org/10.1016/s0009-9236(97)90123-5) [PMID: 9357396].
- [52] Wilson SJ, Rich AS, Rich NC, Potokar J, Nutt DJ. Evaluation of actigraphy and automated telephoned questionnaires to assess hypnotic effects in insomnia. *Int Clin Psychopharmacol* 2004;19(2):77–84. <https://doi.org/10.1097/00004850-200403000-00004> [PMID: 15076015].
- [53] Voshaar RC, van Balkom AJ, Zitman FG. Zolpidem is not superior to temazepam with respect to rebound insomnia: a controlled study. *Eur Neuropsychopharmacol* 2004;14(4):301–6. <https://doi.org/10.1016/j.euro-neuro.2003.09.007> [PMID: 15163440].
- [54] Filligim JM. Double-blind evaluation of the efficacy and safety of temazepam in outpatients with insomnia. *Br J Clin Pharmacol* 1979;8(1):73s–7s. <https://doi.org/10.1111/j.1365-2125.1979.tb00461.x> [PMID: 41546].
- [55] Heffron WA, Roth P. Double-blind evaluation of the safety and hypnotic efficacy of temazepam in insomniac outpatients. *Br J Clin Pharmacol* 1979;8(1):69s–72s. <https://doi.org/10.1111/j.1365-2125.1979.tb00460.x> [PMID: 41544].
- [56] Parsaik AK, Mascarenhas SS, Khosh-Chashm D, Hashmi A, John V, Okusaga O, et al. Mortality associated with anxiolytic and hypnotic drugs-A systematic review and meta-analysis. *Aust N Z J Psychiatr* 2016;50(6):520–33. <https://doi.org/10.1177/0004867415616695> [PMID: 26590022].
- [57] Pisani MA, Murphy TE, Araujo KL, Slattum P, Van Ness PH, Inouye SK. Benzodiazepine and opioid use and the duration of intensive care unit delirium in an older population. *Crit Care Med* 2009;37(1):177–83. <https://doi.org/10.1097/CCM.0b013e318192fcf9> [PMID: 19050611].
- [58] Zaal IJ, Devlin JW, Hazelbag M, Klein Klouwenberg PM, van der Kooi AW, Ong DS, et al. Benzodiazepine-associated delirium in critically ill adults. *Intensive Care Med* 2015;41(12):2130–7. <https://doi.org/10.1007/s00134-015-4063-z> [PMID: 26404392].
- [59] Kok L, Slooter AJ, Hillegers MH, van Dijk D, Veldhuijzen DS. Benzodiazepine use and neuropsychiatric outcomes in the ICU: a systematic review. *Crit Care Med* 2018;46(10):1673–80. <https://doi.org/10.1097/ccm.0000000000000330> [PMID: 29985215].
- [60] Fiest KM, Soo A, Hee Lee C, Niven DJ, Ely EW, Doig CJ, et al. Long-term outcomes in ICU patients with delirium: a population-based cohort study. *Am J Respir Crit Care Med* 2021;204(4):412–20. <https://doi.org/10.1164/rccm.202002-0320OC> [PMID: 33823122].
- [61] Shahid A, Chung SA, Phillipson R, Shapiro CM. An approach to long-term sedative-hypnotic use. *Nat Sci Sleep* 2012;4:53–61. <https://doi.org/10.2147/nss.S28362> [PMID: 23620678].
- [62] Pagel JF, Pandi-Perumal SR, Monti JM. Treating insomnia with medications. *Sleep Sci Pract* 2018;2(1):5. <https://doi.org/10.1186/s41606-018-0025-z> [PMID:].
- [63] Edwards GB, Schuring LM. Pilot study: validating staff nurses' observations of sleep and wake states among critically ill patients, using polysomnography. *Am J Crit Care* 1993;2(2):125–31 [PMID: 8358460].
- [64] Beecroft JM, Ward M, Younes M, Crombach S, Smith O, Hanly PJ. Sleep monitoring in the intensive care unit: comparison of nurse assessment, actigraphy and polysomnography. *Intensive Care Med* 2008;34(11):2076–83. <https://doi.org/10.1007/s00134-008-1180-y> [PMID: 18521566].
- [65] Oxlund J, Knudsen T, Leonthin H, Toft P, Jennum PJ. Subjective sleep assessment compared to polysomnography in mechanically ventilated critically ill ICU patients. *Acta Anaesthesiol Scand* 2023;67(3):311–8. <https://doi.org/10.1111/aas.14190> [PMID: 36576326].
- [66] Benz F, Riemann D, Domschke K, Spiegelhalter K, Johann AF, Marshall NS, et al. How many hours do you sleep? A comparison of subjective and objective sleep duration measures in a sample of insomnia patients and good sleepers. *J Sleep Res* 2023;32(2):e13802. <https://doi.org/10.1111/jsr.13802> [PMID: 36529876].
- [67] Medrzycka-Dabrowska W, Lewandowska K, Kwiecień-Jaguś K, Czyż-Szypanbajl K. Sleep deprivation in intensive care unit - systematic review. *Open Med* 2018;13:384–93 [PMID: 30211321 DOI: 10.1515/med-2018-0057].
- [68] Liu D, Hallt E, Platz A, Humblet A, Lassig-Smith M, Stuart J, et al. Low-dose clonidine infusion to improve sleep in postoperative patients in the high-dependency unit. A randomised placebo-controlled single-centre trial. *Intensive Care Med* 2024;50(11):1873–83 [PMID: 39311905 DOI: 10.1007/s00134-024-07619-w].
- [69] Felten M, Dame C, Lachmann G, Spies C, Rubarth K, Balzer F, et al. Circadian rhythm disruption in critically ill patients. *Acta Physiol* 2023;238(1):e13962. <https://doi.org/10.1111/apha.13962>.
- [70] Prin M, Bertazzo J, Walker LA, Scott B, Eckle T. Enhancing circadian rhythms-the circadian MEGA bundle as novel approach to treat critical illness. *Ann Transl Med* 2023;11(9):319. <https://doi.org/10.21037/atm-22-5127> [PMID: 37404989].
- [71] Nilius G, Richter M, Schroeder M. Updated perspectives on the management of sleep disorders in the intensive care unit. *Nat Sci Sleep* 2021;13:751–62. <https://doi.org/10.2147/nss.S284846> [PMID: 34135650].
- [72] Richards KC, Wang Y-y, Jun J, Ye L. A systematic review of sleep measurement in critically ill patients. *Front Neurol* 2020;11. <https://doi.org/10.3389/fneur.2020.542529> [PMID: 33240191].